# The Bidding Problem as a Bandit: Formulation

Working note

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This note writes the $K$-period bidding problem as one optimization problem, identifies its state, and states the budget constraint in the two forms that matter. The no-learning benchmark, the decomposition (learn, then execute) and the full bandit are the *same* problem with three nested sets of admissible policies.

## 1 Primitives

There are $K$ auctions, $t = 1, \dots, K$. The market state $\theta \in \{\theta_L, \theta_H\}$, $0 < \theta_L < \theta_H$, is drawn once before the first auction with $\Pr(\theta = \theta_H) = \pi_0$ and is never observed; $r := \theta_H/\theta_L$. In auction $t$ the team bids $b_t \geq 0$ and observes $x_t \in \{0, 1\}$ (win or loss), with

$$\Pr(x_t = 1 \mid \theta, b_t) = q_\theta(b_t) := \frac{b_t}{b_t + \theta},$$

independently across auctions given $\theta$ and the bids. A win yields one impression and costs $b_t$; a loss costs nothing. The history after auction $t$ is $H_t = (b_1, x_1, \dots, b_t, x_t)$, with $H_0 = \emptyset$, and $w_t := \sum_{s \leq t} x_s$ is the number of wins so far.

## 2 The problem

The decision variable is not a list of $K$ numbers. It is a *policy* $\beta = (\beta_1, \dots, \beta_K)$ that maps what has been observed to the next bid, $b_t = \beta_t(H_{t-1})$. With a budget $B$ that has to hold in expectation:

$$V(B) = \max_{\beta} \ \mathbb{E}^\beta\Big[\sum_{t=1}^{K} x_t\Big] \qquad \text{subject to} \qquad \mathbb{E}^\beta\Big[\sum_{t=1}^{K} b_t\, x_t\Big] \leq B, \qquad b_t = \beta_t(H_{t-1}) \geq 0. \tag{P}$$

The expectation $\mathbb{E}^\beta$ is taken before the campaign starts: over the state $\theta \sim \pi_0$, the auction outcomes, and any randomisation in the policy.

**Equivalent form.** Let $\pi_t := \Pr(\theta = \theta_H \mid H_t)$ be the belief and $\bar{q}(\pi, b) := \pi q_{\theta_H}(b) + (1 - \pi) q_{\theta_L}(b)$ the predictive win probability. Since $\mathbb{E}[x_t \mid H_{t-1}] = \bar{q}(\pi_{t-1}, b_t)$,

$$V(B) = \max_{\beta} \ \mathbb{E}^\beta\Big[\sum_{t=1}^{K} \bar{q}(\pi_{t-1}, b_t)\Big] \quad \text{s.t.} \quad \mathbb{E}^\beta\Big[\sum_{t=1}^{K} b_t\, \bar{q}(\pi_{t-1}, b_t)\Big] \leq B.$$

## 3 The state: the belief, not the win count

**Bayes' rule.** In log-odds $\ell_t := \log \frac{\pi_t}{1-\pi_t}$, the likelihood ratio of a win at bid $b$ is $q_{\theta_H}(b)/q_{\theta_L}(b) = \frac{b+\theta_L}{b+\theta_H}$ and that of a loss is $r \cdot \frac{b+\theta_L}{b+\theta_H}$. Hence, with $\phi(b) := \log \frac{b+\theta_L}{b+\theta_H} \in (-\log r,\, 0)$,

$$\ell_t = \ell_{t-1} + \phi(b_t) + (1 - x_t)\log r, \qquad \text{so} \qquad \boxed{\ell_t = \ell_0 + \sum_{s \le t} \phi(b_s) + (t - w_t)\log r} \tag{1}$$

Three things follow from (1).

1. *The outcomes enter only through the number of wins $w_t$; the bids enter through $\sum_s \phi(b_s)$, whether they won or not.* A win moves the log-odds by $\phi(b) < 0$, a loss by $\phi(b) + \log r > 0$; the gap between the two is always $\log r$. A low bid makes a win informative and a loss uninformative; a high bid does the opposite.

2. *$(t, w_t)$ is a sufficient statistic only if the bids so far are the same on every history.* That is the case in a probing phase at a fixed bid: with $b_s \equiv b^I = \sqrt{\theta_L \theta_H}$, $\phi(b^I) = -\frac{1}{2}\log r$ and (1) gives the paper's lemma, $\ell_\tau = \ell_0 + \frac{1}{2}(\tau - 2w_\tau)\log r$.

3. *In the full bandit the bids depend on past outcomes, so two histories with the same $(t, w_t)$ carry different beliefs.* Example ($\theta_L = \frac{1}{2}$, $\theta_H = 8$, $\pi_0 = \frac{1}{2}$): bid 2 and win, then bid 1 and lose, gives $\pi_2 = 0.40$; bid 2 and lose, then bid 4 and win, gives $\pi_2 = 0.60$. Both have $t = 2$, $w_2 = 1$.

**Sufficiency of the belief.** For any future bids, the law of future outcomes given $H_t$ depends on $H_t$ only through $\pi_t$: $\Pr(x_{t+1} = 1 \mid H_t, b_{t+1}) = \bar{q}(\pi_t, b_{t+1})$, and $\pi_{t+1}$ is a function of $(\pi_t, b_{t+1}, x_{t+1})$ by (1). Past wins are sunk. So the information state is $(t, \pi_t)$. Whether the remaining budget is also a state depends on the constraint.

## 4 The constraint decides whether the remaining budget is a state

### 4.1 Expected budget (ex ante): state $(t, \pi)$

Problem (P) has a single constraint, so it has a single multiplier $\lambda \ge 0$, the price of a dollar in impressions. For fixed $\lambda$ the problem is an unconstrained dynamic program in which a win at bid $b$ is worth $1 - \lambda b$:

$$J_t^\lambda(\pi) = \max_{b \ge 0} \Big\{ \bar{q}(\pi, b)\big[1 - \lambda b + J_{t+1}^\lambda\big(\pi^+(\pi, b)\big)\big] + \big(1 - \bar{q}(\pi, b)\big) J_{t+1}^\lambda\big(\pi^-(\pi, b)\big) \Big\}, \qquad J_{K+1}^\lambda \equiv 0, \tag{2}$$

where $\pi^+$ and $\pi^-$ are the beliefs after a win and after a loss. Then

$$V(B) = \min_{\lambda \ge 0} \big\{ J_1^\lambda(\pi_0) + \lambda B \big\},$$

and the optimal policy is $b_t = \beta_t^{\lambda^*}(\pi_{t-1})$ at the multiplier whose expected spend equals $B$ (if the spend jumps at $\lambda^*$, randomise between the two policies; with this randomisation there is no duality gap). *The remaining budget is not a state:* how much this particular path has spent does not restrict what it may do next; only the average over paths is constrained, and $\lambda$ carries that.

**How to read (2).** For every fixed policy the Lagrangian value is linear in the prior, so $J_t^\lambda$, a maximum of linear functions, is convex in $\pi$. The belief is a martingale, $\mathbb{E}[\pi' \mid \pi, b] = \pi$, so

$$\underbrace{\bar{q}(\pi, b)\,(1 - \lambda b)}_{\text{immediate value of the bid}} \;+\; J_{t+1}^\lambda(\pi) \;+\; \underbrace{\mathbb{E}\big[J_{t+1}^\lambda(\pi') \mid \pi, b\big] - J_{t+1}^\lambda(\pi)}_{\text{information value of the bid } \geq 0}$$

is what the bid maximises. A *myopic* bidder maximises the first term only; the bandit adds the third. Every bid $b > 0$ is informative here ($q_{\theta_L}(b) \neq q_{\theta_H}(b)$), so the myopic bidder also learns.

### 4.2 Hard budget (every path): state $(t, \pi, R)$

If the realised spend may never exceed $B$,

$$\sum_{t=1}^{K} b_t\, x_t \leq B \;\text{ on every path} \quad\Longleftrightarrow\quad b_t \leq R_{t-1}, \qquad R_t = R_{t-1} - b_t\, x_t, \quad R_0 = B,$$

the remaining budget $R$ becomes a state and the bid is capped by it:

$$V_t(\pi, R) = \max_{0 \leq b \leq R} \Big\{ \bar{q}(\pi, b)\big[1 + V_{t+1}\big(\pi^+(\pi, b), R - b\big)\big] + \big(1 - \bar{q}(\pi, b)\big) V_{t+1}\big(\pi^-(\pi, b), R\big) \Big\}, \qquad V_{K+1} \equiv 0, \tag{3}$$

with value $V_1(\pi_0, B) \leq V(B)$, because every policy that respects the hard budget also respects the expected one.

| Constraint | Formal statement | State |
|---|---|---|
| Expected budget (ex ante) | $\mathbb{E}^\beta[\sum_t b_t x_t] \leq B$ | $(t, \pi_t)$, plus one multiplier $\lambda$ |
| Hard budget | $\sum_t b_t x_t \leq B$ on every path | $(t, \pi_t, R_t)$ |

## 5 Three policy classes in the same problem

Keep the objective and the constraint of (P) and restrict the admissible policies.

**No learning, $\Pi_0$.** The bids are fixed in advance and do not react to outcomes. By concavity the best such plan is a constant bid $b$ with $K\, b\, \bar{q}(\pi_0, b) = B$, which gives the paper's benchmark $W(B) = K\, \bar{q}(\pi_0, b)$.

**Decomposition, $\Pi_D$.** Bid $b^I$ in auctions $1, \ldots, \tau$; then one bid $b(w_\tau)$ in auctions $\tau + 1, \ldots, K$. Here the win count is sufficient (item 2 above), and the problem is

$$V^D(B) = \max_{\tau,\, b(\cdot)} \mathbb{E}\Big[w_\tau + (K - \tau)\, \bar{q}\big(\pi_\tau(w_\tau), b(w_\tau)\big)\Big]$$
$$\text{subject to} \;\; \mathbb{E}\Big[w_\tau b^I + (K - \tau)\, b(w_\tau)\, \bar{q}\big(\pi_\tau(w_\tau), b(w_\tau)\big)\Big] \leq B.$$

**Full bandit, $\Pi_{FB}$.** All policies $b_t = \beta_t(H_{t-1})$; this is (P), solved by (2).

Since $\Pi_0 \subset \Pi_D \subset \Pi_{FB}$, $W(B) \leq V^D(B) \leq V(B)$ for every $B$ and $K$.

*Remark* 1 (The paper's constraint is not the single constraint of (P)). The manuscript requires the exploitation agent to spend its own wallet in expectation after *every* probe outcome: $(K - \tau)\, b(w)\, \bar{q}(\pi_\tau(w), b(w)) = B - w\, b^I$ for each $w = 0, \ldots, \tau$. These $\tau + 1$ constraints imply the single expected-budget constraint (average them over $w$) but are stricter, so the manuscript's value is at most $V^D(B)$. The benchmark $W$, in contrast, faces only the single constraint. To compare the decomposition with the full bandit of (P), both must be given the same constraint.